Exact Finite-Lattice Series and Certified Bounds for the Simple-Cubic Ising Model

Exact 3D Ising Research Program

Draft accompanying the wave-17 artifact bundle

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Abstract

This draft records the current exact-series and rigorous-boundary results for the ferromagnetic nearest-neighbour Ising model on $\mathbb{Z}^3$. Finite-lattice enumeration gives the reduced free-energy series through v^{28} at high temperature and through x^{56} at low temperature, with exact rational coefficients, CRT uniqueness certificates, and independent audits. The coefficients extend the known prefix but do not establish a closed form, D-finiteness, or a thermodynamic-limit solution. A separate exact self-avoiding-walk/inclusion–exclusion certificate improves the lower endpoint of the rigorous critical-coupling interval to $0.21221597532702316272675171 74\ldots$, while the audited upper endpoint remains $0.25273100985866300302600202 66\ldots$. We state the finite combinatorial lemmas, coefficient data, scope of external witnesses, and the unresolved barriers.

1 Model, notation, and evidence tags

Let $K = \beta J$ and $v = \tanh K$. The simple-cubic zero-field model has reduced free energy $\phi(v)$ at high temperature. At low temperature write $x = e^{-2K}$ and use the standard reduced free-energy normalization in the exact finite-lattice method. All displayed coefficients in this draft are exact rational numbers. A finite coefficient computation is denoted [COMPUTATION]; a proved combinatorial implication is [THEOREM] or [LEMMA]; a source used only as an external check is [EXTERNAL].

The repository never uses the benchmark value $K_c = 0.221654626$ to choose a lattice, fit a coefficient, or tune a bound. It is comparison-only.

2 High-temperature series through v^{28}

The finite-lattice method groups connected clusters by their bounding boxes. If a cluster occupies an $a \times b \times c$ box, every edge crossing one of the $a + b + c - 3$ internal cut planes must be paired in the even subgraph. The resulting lower bound on the number of contributing edges gives the exact order filter

$$m \geq 2(a + b + c - 3).$$

Thus the order-28 calculation needs only boxes with $a + b + c \leq 17$. The exact inventory contains 147 canonical permutation classes, with 24 classes on the first $a + b + c = 17$ frontier.

Lemma 2.1 (Finite-lattice order filter)**.** *For a connected even subgraph contained in an $a \times b \times c$ bounding box, the contribution order is at least $2(a+b+c-3)$. Hence all boxes with $a+b+c \geq 18$ are irrelevant through order 28.*

The cap-vector transfer calculation uses exact integer states, profile injections, orbit normalization, and independent CRT uniqueness bounds. The resulting exact coefficient is

$$[v^{28}]\phi(v) = \frac{2102198327465307}{28}, \qquad a_{28} = \frac{525549581866326}{7}, \tag{1}$$

where a_{28} is the associated interaction coefficient in the repository normalization (it is rational, not an integer coefficient). The odd coefficient $[v^{27}]\phi$ vanishes, and every coefficient through v^{26} reproduces the independent predecessor artifact.

This is a finite exact coefficient calculation, not a solution of the 3D model. The cached Arisue–Fujiwara table is a post-computation witness, not an input or proof of the calculation.

3 Low-temperature series through x^{56}

The low-temperature finite-lattice method is organized by a bounding surface. For an $a \times b \times c$ box the minimal surface order is

$$|\partial S| \geq 4(a + b + c) - 6.$$

Consequently the order-56 inventory has $a+b+c \leq 15$ and exactly 455 ordered, or 102 permutation, classes. This inventory is complete through the achieved order x^{56}; by the same surface inequality, the first incomplete inventory is at x^{58}.

For every finite box, the excitation polynomial has nonnegative integer coefficients bounded by the total number of spin configurations. Those coefficient bounds, together with exact CRT reconstruction and per-box uniqueness certificates, prevent an integer coefficient from being inferred from an under-sized modulus.

The exact new coefficients from the audited prefix are:

order	coefficient of the reduced low-temperature series
x^{34}	666750
x^{36}	$-12520405/6$
x^{38}	6583341
x^{40}	$-83409453/4$
x^{42}	$464980286/7$
x^{44}	$-424819905/2$
x^{46}	682202205
x^{48}	$-17588511087/8$
x^{50}	$35552605353/5$
x^{52}	$-46131904167/2$
x^{54}	$675410878105/9$
x^{56}	$-979227570369/4$

All odd powers through x^{56} vanish. In the variable $u = x^2$, the stored prefix reaches u^{28}. The x^{56} result was independently rebuilt with a fresh-prime audit: the producer's exact prefix, 18 survivor certificates, CRT digests, and coefficient normalization all agree.

The external Guttmann–Enting PDF is a post-computation witness for the high-order values. It is not used to select the enumeration or to infer missing coefficients. Five legacy truncation-unobservable rows remain unresolved and require x^{60} or beyond; a producer was stopped because the shared host was saturated, not because an exact coefficient was obtained.

4 Finite structure searches and their limits

The exact coefficient table was also tested against finite ansatz spaces. The representative frontier includes algebraic relations, first- and second-order differential-algebraic relations, Euler ODEs, linear ODEs, and Mahler-type relations. For example, the last frozen high-temperature Euler-ODE row at order two and degree at most six has 21 columns. Its first previously unobservable holdout is v^{24}, where the exact residual is

$$\frac{4997524512213726102393494613686474327029458488 08}{11} \neq 0.$$

Thus that candidate is refuted by its first informative coefficient. The residuals at v^{26} and v^{28} are confirmation only.

The correct conclusion is finite: every searched row is either ruled out by an exact minor or holdout, or remains truncation-unobservable within its prescribed budget. No non-D-finiteness theorem, non-differential-algebraicity theorem, closed form, or relation outside the searched ansatz spaces follows.

5 Certified critical-coupling interval

The current rigorous interval is

$$\boxed{0.21221597532702316272675171742785778 06020 \leq K_c \leq 0.25273100985866300302600202661357012 99926.} \tag{2}$$

The upper endpoint is an audited incumbent from a separate rigorous route; the Peierls head-plus-tail calculation in the present wave does not improve it.

5.1 Lower endpoint from a finite exact union certificate

Let c_{36} be the exact number of length-36 self-avoiding walks on the simple-cubic lattice. A wave-14 inclusion–exclusion union gives the exact producer-certified lower bound

$$a_{35} \geq 1972465461070186257835,$$

and, with the exact c_{36} used by the repository,

$$M = c_{36} - a_{35} = 293939839087363154030 2835.$$

The high-temperature self-avoiding-walk majorant then gives

$$K_c \geq \operatorname{atanh}(M^{-1/36}) = 0.21221597532702316272675171742785778 06020\ldots,$$

with the displayed decimal rounded downward.

The union is composed of pairwise-disjoint height-slab schedules with varying rises, cut positions, and block counts, together with exact defect families and a coordinate-monotone family. All pairwise and triple overlaps in the declared schedule family are evaluated as exact integers. The standalone audit independently re-derives only the declared $\ell \leq 11$, tail ≤ 11 band and the final floor; the full 404-member counts remain producer-certified and are explicitly scoped as such.

A weaker but independently transparent baseline follows from the exact external integer $c_{36} = 2941370856334701726560670$ and submultiplicativity:

$$c_{36q+r} \leq c_{36}^q c_r, \qquad K_c \geq \operatorname{atanh}(c_{36}^{-1/36}) = 0.2122211901166167839331086278 3954\ldots.$$

The stronger displayed endpoint uses the finite inclusion–exclusion improvement, not a fitted connective constant.

5.2 Upper-endpoint scope

The incumbent upper endpoint 0.2527310098586630030260020266... is rigorous in the repository's established bound chain. New exact finite Simon–Lieb prisms, finite-torus infrared/GKS linear programs, and a contour head-plus-tail calculation were audited, but none crosses the incumbent. In particular, the infrared route's two-point class has an $O(1/L)$ floor and the tested Peierls tail converges only well above the target endpoint. These are negative results about named bound families, not an assertion that no sharper inequality exists.

6 Reproducibility and artifact map

The principal exact sources are:

- High temperature: `proofs/ht_v28.md`, `experiments/e94_ht_v28.py`, `results/series/ht_v28.json`, and `tests/test_ht_v28.py`;
- Low temperature: `proofs/lt_x54.md`, `proofs/lt_x56.md`, `experiments/e152_lt_x54.py`, `experiments/e153_lt_x56.py`, `results/series/lt_x34.json`, and `tests/test_lt_x56.py`;
- Structure frontiers: `proofs/series_frontier3.md`, `results/series/frontier3.json`, and `results/series/structure_certificates.json`;
- Critical lower endpoint: `proofs/saw_union4.md`, `experiments/e141_saw_union4.py`, and `results/bounds/saw_union4.json`;
- Rigorous interval and upper-endpoint audit: `proofs/kc_bounds_improved.md`, `proofs/kc_interval2.md`, `proofs/mag_floor.md`, and `results/bounds/`.

The exact producer scripts and standalone tests are included in the bundle. Run from the repository root with `.venv/bin/python`; do not substitute the benchmark coupling for any exact input.

7 Conclusion

The current evidence is a large, exact, independently audited finite prefix and a rigorous but non-sharp critical interval. It advances the boundary of what can be checked exactly without claiming a thermodynamic-limit solution. The next canonical high-temperature coefficient target is v^{30}; the alternate dense FLM route stalls at v^{24}. The low-temperature x^{60} profile is in range but needs a calmer host or a new cut-capped route.